A Comparative Study of U-Net, ResU-Net and Attention U-Net Architectures for Polyp Segmentation of Kvasir-SEG Dataset with Image Preprocessing Pipeline

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Abstract.

Keywords: Computer Aided Diagnosis, Medical Image Processing, Polyps Segmentation, Deep Learning, U-Net.

1 Introduction

Polyps are abnormally developing growths with a potential risk of developing into colorectal cancer (CRC); hence, polyp detection is the key to the prevention and treatment of CRC, which is still a leading cause of cancer-related mortality worldwide [1][2]. Early detection ensures timely removal of polyps, thus preventing their progression towards CRC and considerably reducing mortality. However, there are numerous barriers in the detection of polyps. Polyps are often describable as being small, flat, or resembling surrounding normal tissue, making them difficult to visually detect; in addition, factors such as lack of contrast between the polyp and the tissue, combined with specular reflections from the camera during colonoscopy, complicate their detection. As a result, studies have shown that one in four polyps can be missed [3]. Human factors, including physical fatigue, varying experience, and different judgments further contribute to inconsistent detection rates [2]. It is therefore important to develop Computer Aided Diagnosis (CAD) systems for improving patient outcomes, reducing clinician time, and enhancing diagnostic accuracy.

Polyp segmentation is a basic component of these CAD systems. The task involves pixel-level classification of the polyp region with respect to neighboring healthy tissue. This problem poses serious challenges due to several factors, as indicated earlier. While state-of-the-art architectures such as the FocusU2Net [4], Focus U-Net [5] and nested U-Nets with transformers [6] have set high benchmarks, they often rely on complex designs and high number of parameters. Recent research has demonstrated that achieving high segmentation accuracy on datasets requires not only advanced models but also attention to data quality [7]. This highlights the necessity to improve images through

preprocessing methods and hence improve image quality and highlight relevant features prior to the training of deep learning models.

This study showed that even a light model like U-Net with just 7 million parameters can produce a highly competitive dice score when run using a carefully designed pipeline addressing specific distortions of the image, such as specular reflections and lack of contrast. This result demonstrates an important trade-off: in many cases, a successful preprocessing approach can have as much effect on obtaining improved segmentation accuracy as architectural sophistication.

2 Literature Review

The Polyp segmentation field has been rapidly developing, where deep learning approaches have very often outperformed traditional methods of computer vision. Modern state-of-the-art methods widely rely on the U-Net architecture and enhancements by residual connections, attention mechanisms, and advanced backbones to improve feature learning and cope with problems of diverse polyp shapes and low image contrast. For example, ResUnet++ outperformed traditional UNet and ResUnet architectures, mainly in the case of challenging segmentations, due to the integration of residual connections and attention modules which enhanced feature extraction [8]. Similarly, architectures based on HarDNet backbones combined with multi-scale coding modules and attention mechanisms have demonstrated increased segmentation accuracy, along with reduced computational load, efficiently capturing both global and local features of polyp images [9].

Comparative studies on Kvasir-SEG revealed that architectural improvements like attention gates and ensembling significantly raise the bar [10][11]. Among other things, the literature particularly stresses how image preprocessing is crucial to boosting model performance [12]. Preprocessing techniques, such as correction for specular reflections and enhancing contrast, are crucial in enhancing image quality and segmentation performance by making the polyp boundaries more explicit while reducing visual noise [13].

However, with these developments, there is still a significant lacuna regarding the systematic assessment of the impact of preprocessing on different U-Net architectures. In this study, we proposed a new three-class ground-truth dataset that provides a clinically more appropriate representation of the colonoscopy images by segmenting data into polyp, tissue, and anatomical black background, as shown in **Fig 1**. A comprehensive comparative study has been performed to evaluate the performance of the newly developed dataset and our preprocessing methodology. Models were trained and evaluated three prominent U-Net variants—U-Net [14], ResU-Net [15], and Attention U-Net [16] on three different situations: without preprocessing or postprocessing, with only preprocessing and both preprocessing and postprocessing. This exhaustive analysis, which consists of nine separate experiments, allows us to present a clear-cut

quantitative assessment of the performance gains that can be attributed to the approaches employed and place the findings from this study squarely within the existing body of literature.

3 Methodology

3.1 Dataset Preparation

This research was conducted on publicly shared Kvasir-SEG dataset, consisting of 1000 high-resolution colonoscopy images and their ground-truth masks [17]. We recognized a fundamental limitation in its standard 2-class (polyp or non-polyp) mask representation. This method tends to classify all non-polyp areas as one class, including both the adjacent biological tissue and the dark, non-anatomical periphery of the images.

From the clinical viewpoint, these black areas are not tissue. Categorizing them as such is medically incorrect and can result in erroneous outcomes for Computer-Aided Diagnosis (CAD) systems. Hence, we believe that a stronger segmentation model should distinguish between these different classes. Towards this end, we re-engineered the dataset to 3-class type, including Polyp, Tissue, and Black Background. We employed a Breadth-First Search (BFS) algorithm to detect and rightly reclassify these black areas in the border of our images, so that our new ground-truth masks better reflect actual anatomical tissue. The following figure shows the transition from the initial 2-class ground truth to our updated 3-class version.

(a) (b) (c) (d)

Fig. 1. Comparison of 2-class and 3-class Ground Truth Masks

Fig 1 shows the essential difference between the traditional 2-class ground truth and our improved 3-class segmentation labels. Left to right:

(a) Original Colonoscopy Image: A sample image from the Kvasir-SEG dataset, with a polyp.

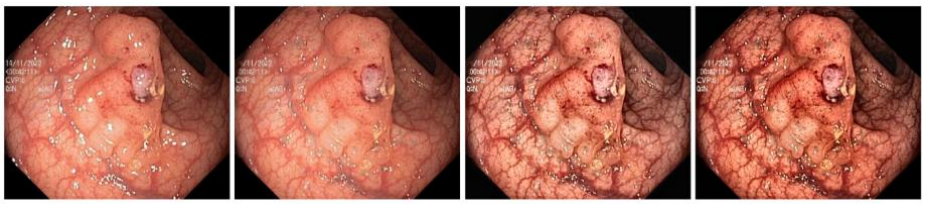
(b) Original 2-Class Ground Truth: The ground truth mask given by the dataset, in which the polyp area is white and all non-polyp regions, including black background and surrounding tissue, are black.

(c) Suggested 3-Class Ground Truth: The proposed ground truth mask, identifying three classes: the polyp (white), the tissue (black), and the non-anatomical background area (gray).

(d) 3-Class Mask Overlay: The three-class mask is overlaid with lower opacity upon the original image to enable meaningful visualization. The polyp is highlighted in red, while the non-anatomical background is highlighted in blue.

3.2 Image Processing

The low-contrast, nonuniform lighting, and specularity reflections in a colonoscopy image were addressed by a multi-step preprocessing pipeline. This pipeline enhances the raw image quality and contributes to the robustness of the trained models. The employed sequence consists of three phases, each of which was designed to mitigate specific image artifacts.



(a) (b) (c) (d)

Fig. 2. Multistage Image Preprocessing Pipeline

Fig 2 illustrates the effect of these below-mentioned steps on the image. From left to right: **(a)** Original Image, showing significant glare and low contrast; **(b)** Specular Reflection Correction, where bright artifacts are successfully removed [18]; **(c)** Contrast Limited Adaptive Histogram Equalization (CLAHE) Application, showing improved local contrast and visibility of subtle features [19]; and **(d)** Gamma Correction, where the overall brightness and intensity are adjusted for a more uniform appearance.

3.3 Model Architecture

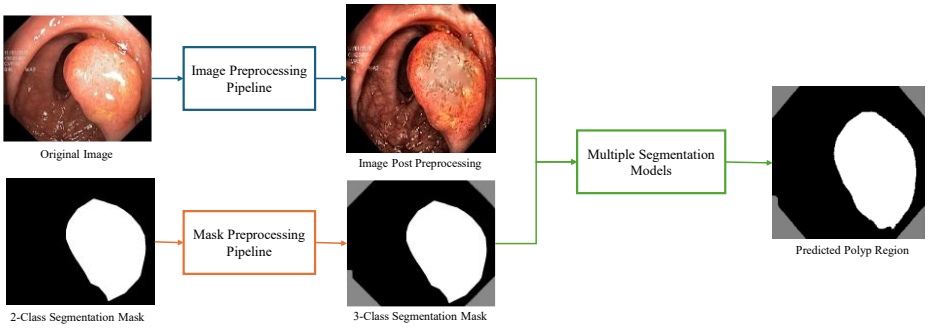


Fig. 3. Illustration of the Polyp Segmentation Process

An overview of the polyp segmentation process has been visually described in **Fig. 3.** The segmentation models in this study are based on the U-Net architecture [14], which is a widely adopted deep learning framework for biomedical image segmentation due to its effective encoder-decoder structure and skip connections. The specific U-Net architecture used for this study is shown in **Fig. 4**.

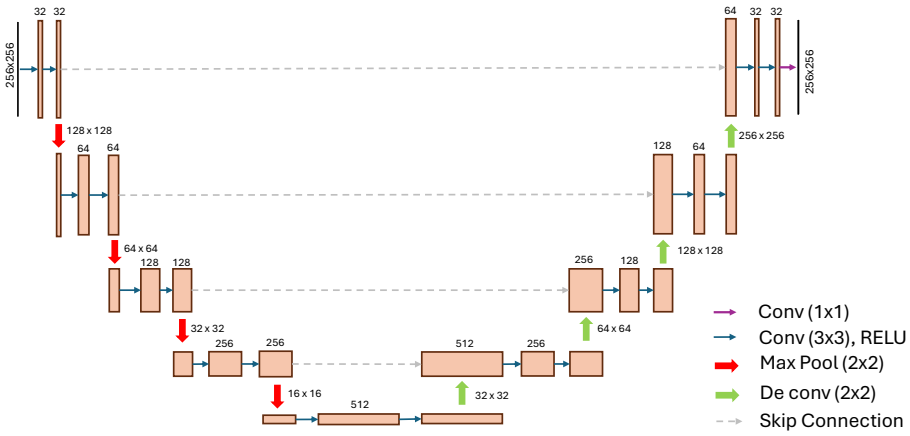


Fig. 4. Baseline U-Net architecture used in this study.

This study specifically evaluated three key variants: the foundational U-Net, the ResU-Net which incorporates residual connections (as shown in **Fig. 5.**) to enhance feature propagation and combat vanishing gradients [15], and the Attention U-Net that uses attention gates (as shown in **Fig. 6.**) to selectively focus on relevant features, improving segmentation accuracy in complex cases [16]. These models were chosen to systematically assess the impact of key architectural enhancements on performance within our proposed data pipeline.

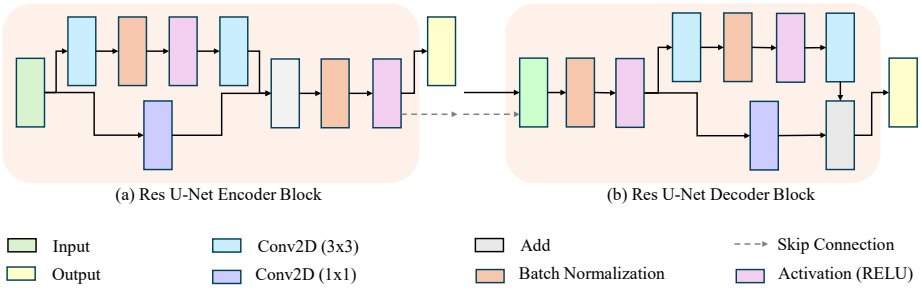


Fig. 5. Encoder and Decoder blocks implemented in Res U-Net

To have a fair evaluation of the three models, the main architecture with each model remains the same with minor changes in each block, **Fig. 5.** Demonstrates the Encoder and Decoder blocks that were implemented in the Res U-Net model trained for this study.

The Attention U-Net architecture incorporates attention gates in between the skip connection and gating signals. **Fig. 6.** represents two inputs, being Input 1 and Input 2 respectively, where Input 1 is the information from the Skip Connection 'x' while Input 2 is the Gating Signal 'g' from the Conv2DTranspose (up-sampled) layer.

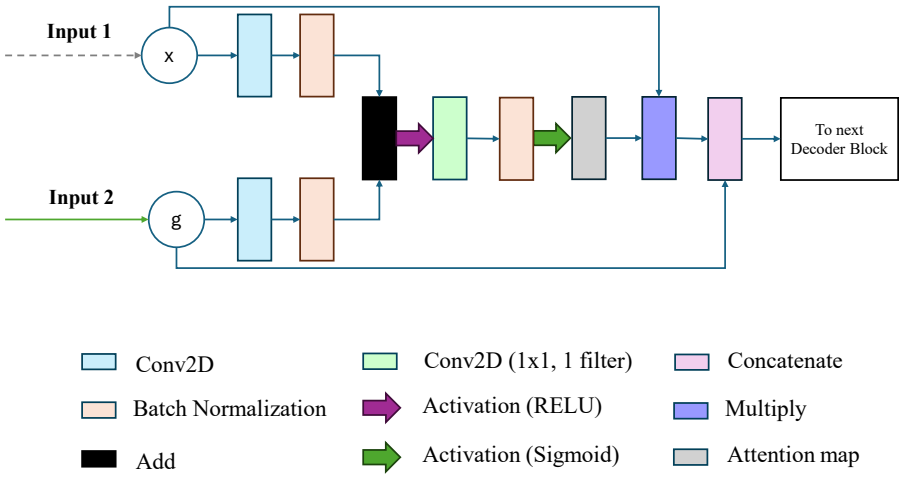


Fig. 6. Attention Gate incorporated in this study

4 Experimental Setup

4.1 Training Environment

All experiments were conducted on a machine with a NVIDIA RTX 3050 Laptop GPU and a Windows 11 operating system. The models were implemented using Python 3.9 and the TensorFlow 2.10 framework. For GPU acceleration, CUDA Toolkit 11.2 and cuDNN 8.1.0 were utilized. The core libraries used for data handling and processing were NumPy 1.26 and OpenCV 4.11.0.86.

4.2 Training Parameters

The training of all three models mentioned in **Section 3.3** was performed using a consistent set of parameters to ensure a fair study. Each model was run for a maximum of 100 epochs, with a batch size of 2. All input images were resized to 256x256 pixels with three color channels. Adam optimizer was used for optimization with an initial learning rate of $1x10^{-4}$. Early stopping with a patience of 15 epochs and Model Checkpoint callbacks were implemented. Learning rate was reduced by a factor of 0.5 if the validation loss of the model did not improve after 7 epochs.

Sparse Categorical Crossentropy was used as the loss function, as it is widely used for multi class segmentation with integer-encoded labels. The loss was calculated using the formula:

$$Loss = \sum_{i=0}^{C-1} y_i \log \hat{y}_i \tag{1}$$

Where C is the number of unique classes, y_i is the binary indicator for class i and $\hat{y}_i$ is the predicted probability of class i.

4.3 Evaluation Metrics

The performance of the models was assessed using the following:

Dice Coefficient.

The measure of similarity between predicted and ground truth masks [20]. The formula for a given class k is:

$$Dice_k = \frac{2 \times \sum_{i=1}^{N} p_{ik} y_{ik}}{\sum_{i=1}^{N} p_{ik} + \sum_{i=1}^{N} y_{ik}} \tag{2}$$

Intersection over Union (IoU).

The Jaccard index (IoU) quantifies the overlap between the predicted and true masks [21]. The formula for a given class k is:

$$IoU_k = \frac{\sum_{i=1}^{N} p_{ik} y_{ik}}{\sum_{i=1}^{N} p_{ik} + \sum_{i=1}^{N} y_{ik} - \sum_{i=1}^{N} p_{ik} y_{ik}} \quad (3)$$

5 Experimental Results

5.1 Quantitative Performance Analysis

Performance of the three variants of U-Net was extensively tested under nine conditions of training, in which the three models were trained on raw and preprocessed images. The models trained on preprocessed images were again tested under two conditions: when an image post-processing pipeline was used and when it was not.

Overall performance measures under all the experimental conditions are presented in **Table 1**. The table indicates consistent improvement in performance because of preprocessing and the additional advantage because of post-processing methods. The 'PP' column signifies application of the preprocessing pipeline, and 'PSP' signifies application of the post-processing on the model output. The performance evaluation metrics employed are the Dice Coefficient and Intersection over Union (IoU) to measure overall segmentation accuracy, and PCP (Polyp Class Precision) and PCR (Polyp Class Recall) for the polyp class in our multi-class segmentation model.

Table 1. Comparative Performance of U-Net Variants with and without Preprocessing and Post-Processing

Model	PP	PSP	Dice Coefficient	IoU	PCP	PCR
U-Net	No	No	0.8833	0.7959	0.8663	0.7603
U-Net	Yes	No	0.9286	0.8706	0.7867	0.8481
U-Net	Yes	Yes	0.9232	0.8611	0.9303	0.6220
ResU-Net	No	No	0.8851	0.7993	0.7683	0.5542
ResU-Net	Yes	No	0.9430	0.8957	0.8769	0.8063
ResU-Net	Yes	Yes	0.9384	0.8877	**0.9347**	0.7190
Attention U-Net	No	No	0.9299	0.8735	0.8724	0.7558
Attention U-Net	Yes	No	**0.9525**	**0.9115**	0.8921	**0.8545**
Attention U-Net	Yes	Yes	0.9514	0.9098	0.9253	0.8143

5.2 Qualitative Results

Fig 7 and **Fig 8** provide a detailed visual analysis of the proposed segmentation approach. **Fig 7** illustrates the end-to-end impact of the preprocessing and post-processing pipelines on the final segmentation output. This high-level comparison is further supported by **Fig 8**, which provides a more granular examination of specific challenging regions, highlighting how each step of the pipeline addresses common image artifacts to improve model performance and accuracy.

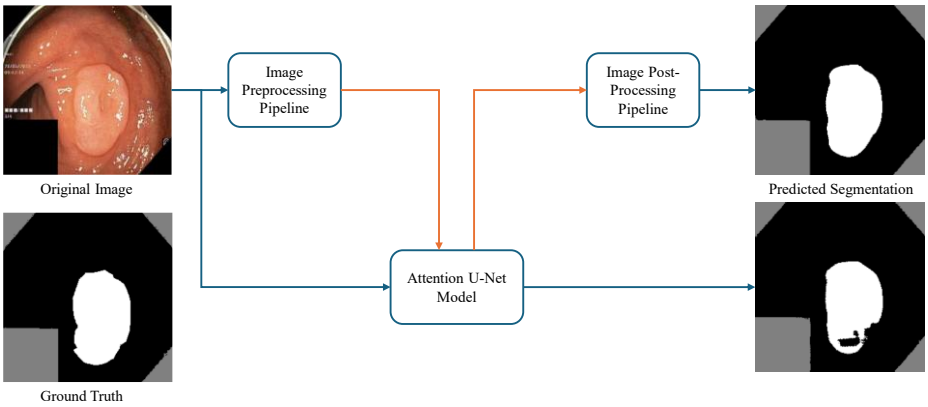
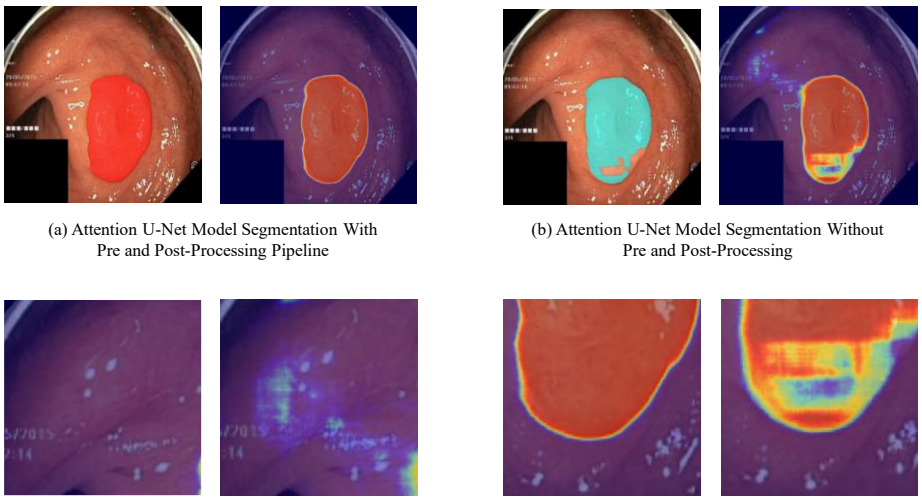


Fig. 7. Illustration of the End-to-End Segmentation Pipelines

Fig 7. presents a block diagram comparing the two main experimental pipelines. The top path shows the flow for the preprocessed and post-processed approach, where an original image is first enhanced before being segmented. The bottom path shows the results from a model trained and tested without these additional steps. The clear difference in the segmented outputs highlights the cumulative performance gain of the proposed pipeline.



(a) Attention U-Net Model Segmentation With Pre and Post-Processing Pipeline

(b) Attention U-Net Model Segmentation Without Pre and Post-Processing

(c) Cropped Region 1 of Grad Cam from (a) and (b)

(d) Cropped Region 2 of Grad Cam from (a) and (b)

Fig. 8. Detailed Qualitative and Experimental Analysis

Fig 8., a multi-panel figure provides an in-depth visual comparison of model behavior and segmentation quality.

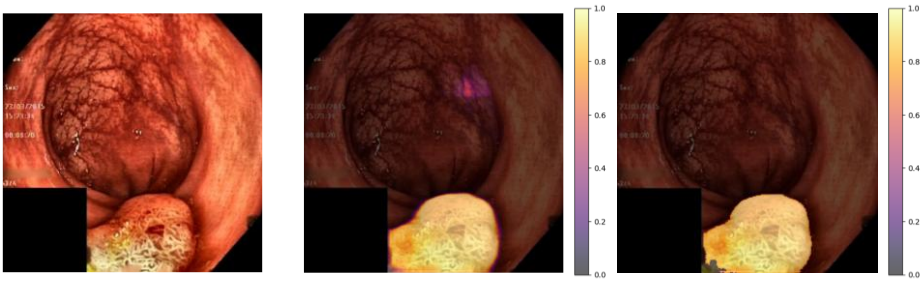
Part (a): This panel shows the final prediction of the preprocessed and post-processed model overlaid on the original image. The accompanying Grad-CAM heatmap from the final layer demonstrates a strong, precise focus on the polyp, indicating effective feature learning.

Part (b): This panel displays the prediction from the model without preprocessing or post-processing overlaid on the original image. The corresponding Grad-CAM heatmap reveals a less focused or more scattered attention pattern, which correlates with the less accurate segmentation.

Part (c): This close-up view focuses on a specular reflection region. A comparative view of the Grad-CAM heatmaps shows the model trained on preprocessed data has no attention on the bright specular reflection. In contrast, the model trained on raw data shows a small amount of focus on this non-anatomical feature, demonstrating the importance of specular reflection correction in training.

Part (d): This panel highlights a difficult-to-segment region of the polyp boundary. The Grad-CAM heatmap for the non-preprocessed model shows a weak and scattered focus, corresponding to a missed prediction. The Grad-CAM for the preprocessed model shows a strong and concentrated focus in the same area, which successfully led to the correct segmentation. This visually demonstrates the critical role of contrast enhancement and gamma correction in defining challenging boundaries for the model.

To demonstrate the effectiveness of the post-processing pipeline employed, we provide a qualitative analysis of the model's confidence maps. This approach visually explains how low-confidence, spurious predictions are filtered out to produce a cleaner, more accurate final mask.



(a) Original Image (b) Confidence Map (Polyp) Before Post-Processing (c) Confidence Map (Polyp) After Post-Processing

Fig. 9. Analysis of Confidence Maps Demonstrating the Impact of Post-Processing

Fig 9. provides a visual breakdown of the post-processing pipeline using confidence maps from the Attention U-Net model. **Part (a)** shows the preprocessed original image.

Part (b) displays the confidence map of the model's raw prediction, where pixel values represent the model's confidence in its classification. Note the presence of a small purple patch, representing low-confidence predictions, in a region that is not a polyp. **Part (c)** shows the confidence map of the final prediction after applying our post-processing pipeline. The absence of the low-confidence purple patch demonstrates how the pipeline successfully filters out spurious predictions, resulting in a more precise and clinically reliable segmentation.

5.3 Comparison with State-of-the-Art Architectures

The enhanced performance of the suggested approach is revealed when contrasted with the current state-of-the-art (SOTA) approaches reported in the literature. The comparison is important to validate the approach and emphasize its contribution. Although the majority of the SOTA approaches address the classical 2-class segmentation task with the classical Kvasir-SEG masks, the suggested approach tackles a more difficult 3-class task with a more informative and clinically useful output.

Detailed Model Analysis.

ResUNet++ [22]: This network, being the direct inspiration behind most modern architectures, was trained on the standard 2-class Kvasir-SEG dataset. It achieved a reported Dice coefficient of 0.8133 and an mIoU of 0.7927. This performance is substantially surpassed by the models presented in this study.

HarDNet-based Model [23]: In this model, regular 2-class segmentation was applied to the Kvasir dataset, and it achieved a Dice of 0.915 and an IoU of 0.862. These are strong results but remain lower than the performance achieved by the present method.

ICSO-AFConvNet [24]: On the Kvasir-SEG 2-class dataset, the model has a Dice of 0.941 and an IoU of 0.906. This is a highly competitive result, but marginally below that of the top-performing model in this study.

PolySeg Plus [25]: This was also evaluated on a 2-class segmentation task on Kvasir-SEG. It achieved a Dice of 0.9476, which showed strong performance.

PNS+ [26]: This model is specifically designed for complex task of video polyp segmentation (VPS). The model had a Dice coefficient of 0.756 on the SUN-SEG dataset, and direct comparison is impossible, but the models of this study are still in a good position.

Table 2. Performance Comparison of the Proposed Model Against State-of-the-Art Approaches

Model	Task	Dice Coefficient	IoU
Our Model	**3-Class Segmentation**	**0.9525**	**0.9115**
PolySeg Plus	2-Class Segmentation	0.9476	-
ICSO-AFConvNet	2-Class Segmentation	0.941	0.906

HarDNet-based Model	2-Class Segmentation	0.915	0.862
ResUNet++	2-Class Segmentation	0.8133	0.7927
PNS+	Video Segmentation	0.756	-

Table 2 gives a clear overview, illustrating how the Attention U-Net model with the suggested preprocessing and post-processing pipeline performs better. The results prove that even when using a multi-class segmentation method, the model performs well, highlighting the clinical benefit of separating relevant tissue from non-anatomical structures like camera obstructions or a narrow field of view.

6 Discussion

The experimental results demonstrate that the proposed pipeline, which integrates preprocessing and post-processing with U-Net variants, is highly effective for polyp segmentation. The quantitative analysis confirmed a consistent performance gain when models were trained on preprocessed data, a finding supported by the qualitative analysis (**Fig 8.**) where Grad-CAM visualizations showed models focusing more precisely on the polyp rather than on image artifacts.

One of the observations from the quantitative analysis is the precision-recall trade-off due to the post-processing pipeline. Although the Dice Coefficient and IoU metrics, being a harmonic mean of the two, saw a marginal drop due to post-processing, a closer look into the polyp class metrics (PCP and PCR) shows a deliberate and visible shift. As shown in Table 1, post-processing, for all the models, dramatically enhanced precision (PCP) and reduced recall (PCR). For instance, the PCP for the U-Net model increased from 0.7867 to 0.9303, showing a dramatic reduction in false positives. In the context of a clinical scenario, such a trade-off can be highly desirable, where the confidence level of the segmentation comes at the expense of the polyp regions identified being highly credible and reducing the likelihood of false alarms leading to unnecessary follow-up interventions.

The comparative performance on the three architectures again demonstrates the superiority of the Attention U-Net. It consistently generated the highest Dice and IoU scores, both with and without post-processing. Its performance can be attributed to the attention mechanisms inherent in it, enabling it to learn more discriminative features and make more consistent, high-confidence predictions. Its drop in Dice score after post-processing by a small extent indicates that its raw output was already better than the others, needing less aggressive filtering. The ResU-Net was also well-performing, reaffirming that residual connections are very helpful for medical image segmentation.

Despite these results, there are some limitations and future work avenues. The models were validated on one, albeit widely used, dataset. Future work can investigate how the pipeline generalizes to other colonoscopy datasets with other camera sensors and image features. Further, exploring the application of this work to real-time video polyp

segmentation is a vital next step, tackling the analysis challenge in dynamic clinical settings.

7 Conclusion

This study presents a holistic polyp segmentation framework that combines a custom preprocessing and post-processing pipeline with well-known variants of U-Net. The proposed methodology addresses salient colonoscopy image challenges, common artifacts and refining model predictions. Experimental results demonstrate significant enhancements in the performance of U-Net, ResU-Net, and Attention U-Net under this framework. In particular, an Attention U-Net model achieved a Dice score of 0.9525 and an IoU of 0.9115, setting a new state-of-the-art benchmark for this segmentation task.

A core contribution of this work is the clinically more relevant three-class segmentation, which our models successfully performed while outperforming existing state-of-the-art models, which were trained on a simpler two-class binary task. Our results highlight that an optimized data pipeline is often more impactful than pure architectural complexity and allow for further development toward more accurate, efficient, and clinically relevant deep learning models in medical image analysis.

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